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ORIGINAL PAPER

Chemical Recycling of Plastic Waste: Comparative Evaluation of Environmental and Economic Performances of Gasification- and Incineration-based Treatment for Lightweight Packaging Waste

Raoul Voss[1,2,3] · Roh Pin Lee[1] · Magnus Fröhling[3]

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Abstract
Chemical recycling (CR) – in enabling the use of plastic waste back as secondary carbon feedstock for production – could play a complementary role to mechanical recycling in supporting the transformation from a linear to a circular carbon economy. To date, research has predominantly focused on assessing technological aspects associated with CR of pure plastic waste streams. Little is known about its potential for treating low-quality and mixed plastic waste fractions which are unsuitable for conventional recycling and are currently incinerated. To address this gap, this investigation utilizes an integrated approach comprising of life cycle assessment and techno-economic analysis to evaluate the environmental and economic performance of CR of lightweight packaging waste via waste gasification compared to direct and indirect incineration in Germany. Results show that CR can contribute significantly – irrespective of the energy mix – to reducing climate change, terrestrial acidification, and fossil resource scarcity. In terms of economic performance, findings suggest that while CR requires higher capital investment, a multi-pronged approach which encompasses upscaling, waiver of carbon dioxide certificate costs, and price premium for CR products could increase profitability of CR to incineration. This study provides empirical support for the potential contribution of CR to complement existing strategies to combat the plastic waste challenge, and insights into market conditions which could promote its economic attractiveness. Additionally, it provides comprehensive inventory data for conventional and alternative waste treatment plants for lightweight packaging waste to inform future research on systemic assessment of CR technologies and their contributions to a circular economy.

Keywords Circular carbon economy · Life cycle assessment (LCA) · Techno-economic analysis (TEA) · Waste incineration · Plastic waste

Abbreviations
AP Acidification potential
CEPCI Chemical Engineering Plant Costs Index

✉ Raoul Voss
raoul-lukas.voss@iec.tu-freiberg.de

Extended author information available on the last page of the article

direct incineration, with the latter being a treatment technique still widely applied in European countries. Incineration reduces waste volume, weight, and pollution potential while generating electricity and/or heat as useful products [18]. Additionally, the extraction of minerals as construction materials and "pure" metals make it an attractive alternative to landfilling [18–22]. However, the downside of incineration is that via the incineration process potentially recyclable carbon-containing LWP fractions such as plastics and cardboard will be converted into greenhouse gas (GHG) emissions. This represents not only a loss of secondary carbon resources which could be channeled back as secondary feedstock for production, it furthermore contributes to global warming [23–25].

Hence, as part of its strategy to support the transition towards a sustainable and efficient economy, the European Union is targeting higher recycling rate of LWP within its New Circular Economy Action Plan [6]. One objective is to further increase the proportion of LWP waste processed in material recovery facilities (MRFs) as a preparatory step for mechanical (material) recycling [7]. Mechanical recycling is defined as methods to convert waste materials into secondary raw materials without changing the basic structure of the material [19, 26]. Through a thorough separation of individual packaging types, MRFs produce homogeneous and high-quality material streams for recyclate production via mechanical recycling. The use of such recyclates for chemical production instead of the direct incineration of LWP waste promotes the transition from a linear economy (i.e. where plastic waste is incinerated following consumption/utilization) to a circular economy (i.e. circularity) where it is channeled back into the production cycle as a secondary feedstock [1, 27]. Additionally, an increase in the use of such high-quality domestic plastic waste streams to replace imported fossil feedstock for packaging production also contributes to primary resource conservation and a reduction of CO_2 leakages along international supply chains [28]. In binding carbon atoms of carbon-containing LWP waste in products rather than in the form of CO_2 emissions after incineration, it furthermore contributes to reducing the carbon footprint of packaging materials and to the achievement of a net-zero GHG emissions society [25, 29].

However, circularity of plastics is easier said than done. Although the EU Directive 94/62/EC states that EU countries must recycle 50% of all LWP made of plastic by 2025 (and 55% until 2030) [7], the example of Germany – a country which despite a modern and well-established plastic waste recycling system in the form of the dual system ("Duales System") [30, 31] only achieves recycling rates of less than 30% for post-consumer plastics [32] – points to the limits of mechanical recycling in meeting the ambitious EU goal for packaging recycling. Thus, to increase the circularity of packaging waste and to meet recycling targets, new recycling concepts and technologies are urgently required to complement mechanical recycling [17]. In this context, chemical recycling as an alternative option to incineration of LWP waste which is not mechanically recyclable has been eliciting increasing attention and interest not only from industry and academics, but also from policymakers and the civil society [8–10, 12, 33–35].

The term chemical recycling (CR) is generally used to describe technologies where carbon-containing waste is broken down into its basic chemical building blocks, which can then be used as raw materials for the production of new consumables such as plastics and chemicals [36, 37]. CR could complement mechanical recycling in increasing the circularity of plastics and supporting the transition from a linear to a circular carbon economy [1, 23, 24, 29, 38, 39]. However, there is a lack of quantitative data and scientific/technical literature to support a comparative evaluation of the strengths and weaknesses of CR against conventional recycling and energy recovery pathways. The predominant attention today – academic and otherwise – is on technical issues related to CR, with existing literature focusing on new technical

developments [10, 40–43], life cycle assessments [23, 44–46], as well as energy and material balances [43, 47, 48]. Furthermore, the predominant focus on pure plastic waste streams as potential CR feedstock has led to significant controversy in the socio-political realm as it presents a direct competition with mechanical recycling techniques [9, 49]. Moreover, despite its potential relevance and contribution to plastic circularity, studies that utilize a multi-dimensional and holistic approach (i.e. considering integration in existing production infrastructures and processes) to comparatively evaluate CR of LWP waste against conventional incineration and mechanical recycling options are, to the best of the authors' knowledge, missing. Such insights would be invaluable in contributing to socio-political discussions about the role of CR in the waste hierarchy, especially for the sustainable management of plastic waste [50].

To address these gaps in the extant literature, this research utilizes a case study approach to investigate quantitative impacts associated with CR of LWP waste against conventional LWP waste treatment pathways. It contributes to discussions about LWP waste treatment through extending the focus from an isolated technical evaluation to integrated environmental and economic assessments of CR in Germany. Specifically, the environmental assessment focuses on a life cycle assessment of indicators namely climate change, terrestrial acidification, and fossil resource scarcity. Economic performance is assessed via techno-economic analysis of total capital investment (TCI), net present value (NPV), dynamic payback period (DPP), and levelized cost of carbon abatement (LCCA). The environmental and economic assessments support a comparative evaluation of three LWP waste treatment pathways namely (1) direct incineration, (2) partial incineration, and (3) CR.

Note that in the investigation, it is assumed that both partial incineration and CR are integrated into current best practices whereby homogeneous and high-quality LWP waste streams are first extracted in MRFs for subsequent mechanical recycling. The heterogeneous and low-quality residues from the MRF sorting process are then either incinerated in the partial incineration pathway, or used as raw material for a CR plant. In this way, CR does not compete with mechanical recycling for pure plastic waste streams. Rather, it complements mechanical recycling to raise the recycling quota by presenting a recycling alternative to incineration for residue LWP waste fractions which are not suitable for mechanical recycling [51].

The paper is structured as follows: the "Study Context" section provides detailed information on the study context in Germany. This supports an assessment of the transferability of findings, conclusions, and recommendations of the current study to other research contexts, e.g. to other countries with different waste management systems. The "Methodology" section presents the inventory dataset for this study. In addition to an overview of the characteristics of LWP waste and treatment technology, methodologies utilized for the environmental and economic calculations are also presented. The "Results and Discussion" section presents the results and discusses the comparative evaluation of the three different LWP waste treatment pathways in terms of their environmental and economic performance. Finally, the paper concludes with a summary of study implications and a discussion about its limitations and transferability to other contexts or subsequent studies.

Study Context

With its dual system, Germany has a structured source separation system in place where LWP waste is collected separately from other waste (i.e. residual municipal solid waste, paper/cardboard waste, glass bottles, etc.) in yellow containers and sacks ("Gelbe Tonne" and "Gelber Sack") [51]. The systems implementation was motivated by the enactment of

the "Packaging Ordinance" in the early 1990s which anchored the "polluter pays principle" in Germany's waste management legislation [52]. Since then, companies that introduce LWPs to the market are obliged to contribute financially to their collection and recovery. Today, more than 2 million tonnes of LWP waste are collected annually by the dual system and are processed in MRFs [53].

The thorough separation of the individual packaging types in MRFs facilitates the production of homogeneous and high-quality material streams for recycling. However, the degree of separation depends on the efficiency of the sorting process. Inevitably, MRFs produce a residual fraction of materials (i.e. sorting residues) that are either too small for sorting or consist of multiple partially bonded materials that do not meet the quality standards for subsequent mechanical recycling processes [13]. According to the German Nature and Biodiversity Conservation Union (NABU), more than 50% of LWP waste ends up as sorting residues which are not suitable for mechanical recycling [54]. Currently, these residues are typically processed as refuse-derived fuel (RDF) for energy and/or heat recovery in incineration facilities ranging from RDF power plants, cement works, to coal-fired power plants [51]. However, as sorting residues from LWP waste are carbon-rich, a potential alternative treatment pathway to incineration is the utilization of LWP sorting residues as secondary carbon feedstock for chemical production via CR.

Since the 1970s, a keen interest in CR development is evident from the deployment of a variety of experimental-, pilot-, and industrial-scale facilities in different countries [10, 55]. The term CR generally refers to the decomposition of carbonaceous waste either thermally or by solvents into synthesis gas monomers, oligomers, or higher hydrocarbons as products [10]. These products represent feedstocks for the production of plastics or other value products such as fertilizers, paints, or pharmaceuticals [8, 37, 56, 57]. According to Mamani et al. [10], CR technologies can be generally categorized in solvolysis, pyrolysis, and gasification [10]. However, not all CR technologies are suitable for processing LWP sorting residues with mixed compositions and low qualities. To date, only gasification technologies have been operated on an industrial-scale for such purpose. For example, in Berrenrath from 1993 to 1997 and in Schwarze Pumpe from 1995 to 2007 [40, 55, 58, 59], LWP sorting residues from the dual system served as co-feedstock with coal for syngas generation and subsequent methanol production in Germany. Another prominent example is Showa Denko K.K. in Kawasaki/Japan, where a plastic waste gasification plant has been in operation since 2003 to generate hydrogen for ammonia production [60].

In view of its previous and current practical application for the CR of plastic waste, this research evaluates gasification as the alternative CR treatment pathway to incineration pathways for LWP waste. Unlike incineration pathways where electricity and heat are the desired products, during the gasification process waste is broken down into a CO- and H_2-rich syngas under high pressure and high temperature conditions with oxygen and/or steam as gasification agents. In subsequent downstream processes, syngas can be converted to platform chemicals such as methanol, which can then be further synthesized via a methanol-to-olefins process to olefins, an important intermediate raw material for the production of a broad spectrum of chemical products including virgin plastics [25, 61, 62].

Figure 1 illustrates the three LWP waste treatment pathways considered in this study. In Pathway I (i.e. reference scenario), LWP waste is channeled directly for incineration in RDF power plants without pre-sorting (i.e. no material recovery in MRF). Note that this does not reflect the case in Germany as LWP waste is pre-sorted in MRFs for mechanical recycling before sorting residues are sent as RDF for incineration (i.e. partial incineration). Nevertheless, in view that incineration is widely practiced and/or considered for deployment in numerous EU countries (and worldwide), the direct

Inventory Dataset

Definition and Characterization of LWP Waste in Germany

For the investigation, LWP waste is defined as waste material that is assigned to the waste code number EAV-15010601-U for "light weight packaging" in the disposal statistics by the German Federal Statistical Office [53]. Note that German authorities utilize EAV-15010601-U as a subgroup for the European waste code number EAV-150106 for “mixed packaging” [53, 63]. The assumed LWP waste composition as presented in Table S1 in the Supplementary Information (SI) relies on data from a study conducted by the German Environment Agency (UBA) [64]. Chemical composition of each subfraction is based on Riber et al. [65], resulting in chemical characteristics for LWP waste – e.g. water content, lower calorific value, and carbon content – as presented in Table S2 in the SI.

Technological System

Scientific publications, governmental technical reports, LCI databases, and process modeling results serve as data sources for the investigated treatment technologies. Specifically, data on waste collection and waste transportation is drawn from Larsen [66] and Olesen [67]. Data for waste incineration is obtained from technical reports by UBA [51, 68] and corresponding BAT-documents by the European Commission [69, 70]. Data on materials recovery is drawn from research by Cimpan et al. [13]. Finally, data on chemical recycling is obtained from chemical process modeling via ASPEN Plus V10 based on the approach by Keller et al. [23]. Detailed material flow assumptions are presented in the following paragraphs. Aggregated inputs and outputs for each treatment pathway are illustrated in Fig. 1.

Collection and transportation LWP waste collection is modeled assuming the combustion of 3.1 L diesel tonne^{-1} in a waste collection truck [66]. Transportation of waste treatment outputs such as bottom ash, fly ash, slag, and sorting products is modeled under the assumption of a Euro-5 semitrailer truck with a cargo volume of 10 tonnes travelling fully loaded, consuming 0.021 L diesel tonne^{-1} km^{-1} [67].

Waste incineration Incineration of LWP waste and LWP waste sorting residues is based on RDF incineration in German RDF power plants as described by UBA [51, 68]. The process consists of four steps, namely 1) delivery and storage, 2) grate-firing, 3) flue gas cleaning, and 4) metal separation:

- Step 1: Delivery and storage of LWP waste after collection are modeled via the combustion of 0.5 L diesel tonne^{-1} in a wheel loader that operates at 33% average load factor [71]. Note that waste incineration does not include delivery and storage in PII, as it describes the second treatment step after materials recovery (cf. Fig. 1).
- Step 2: Grate-firing consumes 2 L fuel oil tonne^{-1} input as auxiliary fuel [70] and cogenerates heat and electricity. Cogeneration efficiencies are set to 37% for heat and 15% for electricity relative to the lower heating value of the input [68]. Please note that assumed values correspond to net values i.e. after deduction of internal energy consumption. Excess heat and electricity are fed into a local heat grid and the

German electricity grid, respectively, to substitute conventional energy. Bottom ash from grate-firing is processed in Step 4.

- Step 3: Gas treatment for airborne emissions resulting from grate-firing is assumed with a wet flue gas cleaning based on data provided by UBA [51] and EC [70]. Specifically, the system consists of a dry electrostatic precipitator, a two-stage water scrubber unit, a selective catalytic reforming unit, and a fly-flow adsorption. Supply consumption is set to 0.8 kg activated carbon, 5.9 kg ammonia, 12 kg calcium carbonate, and 3 kg sodium hydroxide tonne^{-1} input [51, 70].
- Step 4: A metal separation unit – comprising of two overbelt magnets, a bar sizer, an impact crusher, a flip-flop sieve, and an eddy current separator – with recovery efficiencies of 80% for ferrous and 19.5% for non-ferrous metals is assumed [72].

Process-specific air emission of treatment include 750 g NOx (4.3 g N_2O), 260 g CO, 150 g SO_2,, and 2.5 g CH_4 tonne^{-1} input [73, 74]. Fossil CO_2 emission are calculated based on the fossil carbon content of input in EASETECH (cf. "Life Cycle Assessment Model"). Solid emissions include fly ash and bottom ash which are treated as follows: Fly ash is assumed to be transported 700 km to a salt mine (cf. process "collection and transportation") where it is utilized as backfilling material consuming 1.8 L diesel and 28 kWh electricity tonne^{-1} fly ash [75]. Remaining bottom ash after metal separation is transported 2 km to a landfill, modeled with the combustion of 1 L diesel tonne^{-1} in a 89 KW drivable machine charged to 36% on average [76]. Note that transportation via truck is generally considered one-way due to the assumption of optimized logistics.

Materials recovery Materials recovery is modeled based on data provided by Cimpan et al. [13] for MRFs in Germany. Treatment consists of two treatment steps, namely 1) delivery and storage and 2) material sorting:

- Step 1: Delivery and storage include reception of LWP waste and a pre-conditioning in form of bag opening and size reduction consuming 0.84 L diesel and 4.1 kWh electricity tonne^{-1} LWP waste input [13].
- Step 2: Material sorting consists of i) a conditioning step with sieving steps or ballistic separation, ii) a primary sorting step for first separation into material groups, iii) a refining step to increase material output quality i.a. via manual sorting, and iv) a product handling step to prepare products for downstream processors via e.g. balling [13]. Assumed sorting efficiencies are based on data provided by Cimpan et al. [13] and UBA [51] for materials recovery processes with extensive plastic sorting (cf. SI, Table S3). Supply consumption includes 1.4 L diesel and 87 kWh electricity tonne^{-1} LWP waste input [13].

Material recovery products include i) recyclable materials, ii) sorting residues, and iii) mixed plastics. For i), recyclable materials such as packaging polymers, plastic foil, and metals are transported 20 km to a recycling facility [13]. For ii), sorting residues are transferred to the next treatment step i.e. waste incineration or CR assuming the combustion of 0.5 L diesel tonne^{-1} in a wheel loader that operates at 33% average load factor [71]. For iii), mixed plastics are assumed to be treated via incineration based on Cimpan et al. [13]. As PII includes a corresponding treatment step, mixed plastics are assumed to be handled together with sorting residues (cf. Fig. 1). As PIII does not include a corresponding treatment step, mixed plastics are transported 20 km to an external RDF power plant assuming

ery stage [23]. For i) syngas-to-methanol synthesis, a single stage quasi-isothermal fixed bed reactor of the Lurgi type is assumed. For ii) methanol-to-olefins synthesis, assumed process configuration refers to the UOP HYDRO methanol-to-olefins process [23]. In iii) the olefins recovery stage, product gas obtained from methanol-to-olefins process is introduced to an olefin recovery process that includes a sodium hydroxide scrubber, a cryogenic separation unit, a three-stage refringent plant, and an adsorption & extractive distillation unit to recover butene and butadiene.

Note that data for treatment steps 2) gasification, 3) gas treatment, and 4) synthesis is produced with the chemical process modeling software ASPEN Plus V10 as presented by Keller et al. [23]. Aggregated supply consumption includes 280 kWh electricity and 9.1 kg sodium hydroxide tonne^{-1} sorting residue input. Treatment products of the CR process are assumed with 4.2 kg butadiene, 1.9 kg butene, 0.76 kg fuel gas, 710 kWh heat as process steam that can be used for versatile applications at the plant site and/or for export, 16 kg liquified petroleum gas, and 240 kg olefins (120 kg ethylene, 130 kg propylene) tonne^{-1} sorting residue input. Airborne emissions are accounted for with 710 kg fossil CO_2, 0.042 kg CO, 0.13 kg SO_2, and 0.1 kg NOx tonne^{-1} sorting residue input. 99 kg slag tonne^{-1} sorting residue input are transported 2 km to an adjacent landfill where it is deposited, modeled with the combustion of 1 L diesel tonne^{-1} in a 89 KW drivable machine charged to 36% on average [76].

Environmental Assessment

ISO standard 14,040:2006 which defines life cycle assessment requirements is utilized to assess the environmental life cycle impacts of all treatment pathways [87].

Goal and Scope

Goal is to facilitate a comparative assessment of environmental impacts associated with the three pathways for LWP waste treatment, namely direct incineration in a RDF power plant without pre-sorting, incineration of sorting residues from MRF, and CR of sorting residues from MRF. The functional unit is defined as the treatment of one metric tonne of LWP waste, with fractional composition and chemical characteristics further described in Tables S1 and S2 in the SI. The geographical scope is Germany and the temporal scope is set to 2021 until 2028 as assumptions regarding waste composition or technology efficiency are based on current knowledge [79].

System Boundaries

As illustrated in Fig. 1, system boundaries extend from the point of waste collection in households to treatment product distribution. Waste generation upstream impacts are excluded as a zero-burden approach is applied [79]. Nevertheless, upstream impacts associated with supply utilization and downstream impacts associated with treatment product distribution are included via system expansion with a background system (cf. Table 1). Corresponding environmental footprints for supplies are drawn directly from GaBi [77]. In contrast, footprints for treatment products are calculated based on data provided by Montejo et al. [79] for market product substitution ratios that reflect quality/material losses

during recycling processes and reduced market acceptance. Specifically, ferrous and non-ferrous scraps substitute primary steel and aluminum at substitution ratios of 87% and 79%, respectively. Paper & cardboard waste substitutes primary paper at a ratio of 82%. Both packaging polymers and plastic foil substitute primary low-density polyethylene at a substitution ratio of 81%. Compounds are assumed to contain 74% paper, 22% plastic foil, and 4% aluminum based on the material composition of beverage cartons [79]. Material loss during decomposition is set to 22%. After decomposition, individual components substitute primary market products at ratios defined above. Eventually, chemical recycling products, electricity, and heat substitute market products at a ratio of 100% as conventional quality can be expected [46]. Footprints for supplies and market products are drawn from GaBi [77]. Note that all applied footprints refer to specific or market average values, indicating the attributional mode of this assessment. Moreover, environmental impacts associated with plant construction are beyond the scope of the current investigation.

Life Cycle Assessment Model

The software EASETEACH V3.1.7. [88] – well established for waste management research [89–91] – is applied for the environmental assessment. EASETECH is developed at the Department of Environmental Engineering, Technical University of Denmark (DTU), and focuses i.a. on a detailed consideration of fractional composition and chemical characteristics of waste. In EASETECH, LWP waste can be modeled by combining predefined sub-fractions such as standard packaging polymers, plastic foils, and paper waste. Based on the assumed composition, the software then determines key characteristics of LWP waste including energy content, carbon content, and water content (cf. SI, Table S2 for chemical characteristics). At the same time, it allows a realistic representation of sorting processes as individual sorting efficiencies can be assumed for different sub-fractions.

Impact Assessment

Keller et al. [23] identified midpoint indicators climate change, terrestrial acidification, and fossil resource scarcity as relevant impact categories for the examination of incineration and CR via gasification for alternative carbon carriers in Germany. Note that further life cycle impact categories used in holistic analyses such as marine eutrophication or land use are not analyzed in the current investigation. Regarding impact assessment methods, impact on climate change is determined per mechanism global warming potential for a period of 100 year (GWP100) in line with the Fifth Assessment Report (AR5) of IPCC [92]. Biogenic CO_2 emissions are considered climate-neutral in the investigation. Impact on terrestrial acidification is determined per mechanism acidification potential (AP) and impact on fossil resource scarcity per mechanism fossil fuel potential (FFP) based on the ReCiPe2016 framework by Huijbregts et al. [93].

Environmental Sensitivity Analysis

As the German energy system is currently undergoing substantial transformations [94], different compositions for energy supply, i.e. energy provision scenarios (EPS), are examined in more detail. Via sensitivity analysis, using data provided by the European Commission [95] for electricity provision and by BDI [81] for heat provision, the influence of changes

Environmental Impacts

Figure 2a–c present results from climate change, terrestrial acidification, and fossil resource scarcity assessments, respectively. For the analysis, impacts are divided into 1) process impacts, 2) heat substitution, 3) electricity substitution, 4) ferrous metal substitution, 5) non-ferrous metal substitution, 6) recyclables recovery, and 7) CR products recovery. Results of the contribution analysis indicate that heat and electricity substitution have a major effect on the environmental viability of all impact categories. In the following paragraphs, results for the investigated scenarios will be discussed for each individual impact category.

Climate Change

In scenario EPS-BASIC, PI exhibits the highest GWP100 at 290 kg CO_2eq tonne^{-1} LWP waste, whereas PII and PIII are associated with a GWP100 reduction at -110 and -150 kg CO_2eq tonne^{-1} LWP waste, respectively. The high GWP100 reduction of PIII is due to low process emissions and chemical product substitution. In scenario EPS-PRED, assuming a predominant renewable energy supply, both PI and PII produce net positive CO_2eq. emissions at 700 and 200 kg CO_2eq tonne^{-1} LWP waste, respectively, as emission credits for the substitution of electricity and heat are significantly reduced. In contrast, GWP100 reduction associated with PIII remains similar to that in scenario EPS-BASIC. In scenario EPS-CN with a climate neutral energy supply, while the GWP100 reduction effect of PIII is reduced by 50% to -100 kg CO_2eq tonne^{-1} LWP waste, the GWP100 of PI and PII will further increase to 1200 and 580 kg CO_2eq tonne^{-1} LWP waste, respectively. Results thus strongly point to the GWP reduction potential of CR-based pathway PIII – irrespective of energy systems – in reducing carbon emissions associated with LWP waste treatment in comparison to incineration-based pathways PI and PII.

Terrestrial Acidification

Across all scenarios, PIII maintains a consistent contribution to reducing AP as a measure for the impact on terrestrial acidification at -1.4 to -1.3 kg SO_2eq. tonne^{-1} LWP waste. This is because the lower credits for heat substitution in scenarios with increasing renewable energy supply are balanced with lower indirect SO_2eq. emissions that are induced by electricity consumption. In contrast, while incineration-based pathways also contribute to reducing AP, their contribution, in particularly for PI, is dependent on the energy supply in Germany. Hence, as process emissions and emission credits from electricity production become increasingly reduced corresponding to an increasing share of renewable energy supply, AP reduction also decreases considerably. In the case of PI, it reduces from -0.61 in Scenario BASIC to -0.34, and 0.075 kg SO_2eq. tonne^{-1} LWP waste. Taken together, while all treatment pathways contribute to a AP reduction, the contribution of CR pathway is largely independent of energy supply and remains consistent even assuming renewable energy supply. In contrast, the contribution of incineration-based pathways to AP reduction is dependent on the energy mix and exhibits a tendency to decrease as the share of renewables in the energy mix increases.

Fossil Resource Scarcity

Generally, all pathways contribute to a reduction in FFP as measure for the impact on fossil resource scarcity. For the incineration-based pathways PI and PII, this is primarily due to

Dynamic Payback Period (DPP)

As illustrated in Fig. 4b, DPP is only relevant for pathways with positive NPV (i.e. investment will be repaid within the investment timeline). Under scenario ES-BASIC, direct incineration with a DPP of 8 years exhibits a better project feasibility than indirect incineration with a DPP of 11 years. With upscaling of plant capacity, the project attractivity of incineration-based pathways improves with a reduced DPP of 6 years. In contrast, the CR pathway will not be as economically attractive as incineration-based pathways in terms of DPP as it requires twice as long for the investment to be repaid. Nevertheless, even with the longer DPP, the investment will be paid back in less than half of the investment lifespan of the CR plant (i.e. at 33 years). Not surprisingly, environmental regulation relating to CO_2 certificate costs for CR is observed to have no impact on DPP for incineration-based pathways compared to scenario ES-BASIC. Assuming a price premium for CR products, a similar DPP for PIII (i.e. 9 years) and PII (i.e. 8 years) is observed.

Levelized Costs of Carbon Abatement (LCCA)

LCCA is calculated for PII and PIII using PI as the reference pathway. The energy mix is observed to exert a significant impact on carbon emissions for LWP waste treatment pathways (see the "Climate Change" section). This will have a corresponding impact on LCCA. The difference in energy mixes is thus accounted for in the LCCA via the energy scenarios EPS-BASIC, EPS-PRED, and EPS-CN.

Not surprisingly, for all economic scenarios (i.e. ES-BASIC, ES-PS, ES-EREG, and ES-MC), LCCA for both indirect incineration (i.e. PII) and CR (i.e. PIII) – compared to PI as reference pathway – will decrease as the proportion of renewables in the energy mix increases (see Fig. 5). Interestingly, under existing market conditions i.e. ES-BASIC, while PII will incur lower LCCA compared to PIII under EPS-BASIC condition (i.e. 130 compared to 180 € $tonne^{-1}$ CO_2 eq.), this tendency is reversed as the proportion of renewables in the energy mix increases. Specifically, PIII incurs less LCCA compared to PII in EPS-PRED and EPS-CN at 93 and 62 € $tonne^{-1}$ CO_2 eq. compared to 110 and 89 € $tonne^{-1}$ CO_2 eq., respectively. This indicates that with increasing renewables in the energy mix, the attractivity of CR compared to indirect incineration (i.e. in terms of reduced LCCA) will correspondingly increase.

For PII, the only economic scenario which has an impact in reducing its LCCA is ES-PS. For PIII, not only do plant scaling (ES-PS) and a waiver of CO_2 certificate costs for CR (ES-EREG) lead to a significant reduction of LCCA; in the case of a price premium for CR products (ES-MC), negative LCCA values indicate that carbon saving and cost reduction can be attained at the same time.

Taken together, assuming an increasing proportion of renewable energy supply in the future as economies strive for carbon neutrality and net-zero, CR will present an economically attractive alternative to indirect incineration to abate carbon emissions across diverse economic scenarios.

Conclusion

This study quantitatively evaluates the environmental and economic impacts associated with gasification as an alternative chemical recycling treatment pathway to direct and indirect incineration pathways for lightweight packaging waste. To support the comparative